

Overview of Vanishing gradient problem

Subject: Vanishing gradient problem

1.

Geometric model Continue using the above one-neuron network, fixing $w = 5$, $x(0) = 0.5$, $u(t) = 0$ $\{\displaystyle w=5, x(0)=0.5, u(t)=0\}$, and consider a loss function defined by $L(x(T)) = (0.855 - x(T))^2$ $\{\displaystyle L(x(T))=(0.855-x(T))^2\}$. This produces a rather pathological loss landscape: as b $\{\displaystyle b\}$ approach -2.5 $\{\displaystyle -2.5\}$ from above, the loss approaches zero, but as soon as b $\{\displaystyle b\}$ crosses -2.5 $\{\displaystyle -2.5\}$, the attractor basin changes, and loss jumps to 0.50. Consequently, attempting to train b $\{\displaystyle b\}$ by gradient descent would "hit a wall in the loss landscape", and cause exploding gradient. A slightly more complex situation is plotted in, Figures 6.

2.

Since $|\sigma'| \leq 1$ $\{\displaystyle \left|\sigma'\right|\leq 1\}$, the operator norm of the above multiplication is bounded above by $\|W_{\text{rec}}\|^k$ $\{\displaystyle \left\|W_{\text{rec}}\right\|^k\}$. So if the spectral radius of W_{rec} $\{\displaystyle W_{\text{rec}}\}$ is $\gamma < 1$ $\{\displaystyle \gamma < 1\}$, then at large k $\{\displaystyle k\}$, the above multiplication has operator norm bounded above by $\gamma^k \rightarrow 0$ $\{\displaystyle \gamma^k \rightarrow 0\}$. This is the prototypical vanishing gradient problem. The effect of a vanishing gradient is that the network cannot learn long-range effects. Recall Equation (loss differential): $\nabla_{\theta} L = \nabla_x L(x_T, u_1, \dots, u_T) [\nabla_{\theta} F(x_{t-1}, u_t, \theta) + \nabla_x F(x_{t-1}, u_t, \theta) \nabla_{\theta} F(x_{t-2}, u_{t-1}, \theta) + \dots]$ $\{\displaystyle \nabla_{\theta} L = \nabla_x L(x_T, u_1, \dots, u_T) [\nabla_{\theta} F(x_{t-1}, u_t, \theta) + \nabla_x F(x_{t-1}, u_t, \theta) \nabla_{\theta} F(x_{t-2}, u_{t-1}, \theta) + \dots]\}$. The components of $\nabla_{\theta} F(x, u, \theta)$ $\{\displaystyle \nabla_{\theta} F(x, u, \theta)\}$ are just components of $\sigma(x)$ $\{\displaystyle \sigma(x)\}$ and u $\{\displaystyle u\}$, so if $u_t, u_{t-1}, \dots$ $\{\displaystyle u_t, u_{t-1}, \dots\}$ are bounded, then $\|\nabla_{\theta} F(x_{t-k-1}, u_{t-k}, \theta)\|$ $\{\displaystyle \left\|\nabla_{\theta} F(x_{t-k-1}, u_{t-k}, \theta)\right\|\}$ is also bounded by some $M > 0$ $\{\displaystyle M > 0\}$, and so the terms in $\nabla_{\theta} L$ $\{\displaystyle \nabla_{\theta} L\}$ decay as $M \gamma^k$ $\{\displaystyle M \gamma^k\}$. This means that, effectively, $\nabla_{\theta} L$ $\{\displaystyle \nabla_{\theta} L\}$ is affected only by the first $O(\gamma^{-1})$ $\{\displaystyle O(\gamma^{-1})\}$ terms in the sum. If $\gamma \geq 1$ $\{\displaystyle \gamma \geq 1\}$, the above analysis does not quite work. For the prototypical exploding gradient problem, the next model is clearer.

3.

Consider first the autonomous case, with $u = 0$ $\{\displaystyle u=0\}$. Set $w = 5.0$ $\{\displaystyle w=5.0\}$, and vary b $\{\displaystyle b\}$ in $[-3, -2]$ $\{\displaystyle [-3, -2]\}$. As b $\{\displaystyle b\}$ decreases, the system has 1 stable point, then has 2 stable points and 1 unstable point, and finally has 1 stable point again. Explicitly, the stable points are $(x, b) = (x, \ln(\frac{x}{1-x}) - 5x)$ $\{\displaystyle (x, b) = (x, \ln(\frac{x}{1-x}) - 5x)\}$. Now consider $\frac{\Delta x(T)}{\Delta x(0)}$ $\{\displaystyle \frac{\Delta x(T)}{\Delta x(0)}\}$ and $\frac{\Delta x(T)}{\Delta b}$ $\{\displaystyle \frac{\Delta x(T)}{\Delta b}\}$, where T $\{\displaystyle T\}$ is large enough that the system has settled into one of the stable points. If $(x(0), b)$ $\{\displaystyle (x(0), b)\}$ puts the system very close to an unstable point, then a tiny variation in $x(0)$ $\{\displaystyle x(0)\}$ or b $\{\displaystyle b\}$ would make $x(T)$ $\{\displaystyle x(T)\}$ move from

one stable point to the other. This makes $\frac{\Delta x(T)}{\Delta x(0)}$ and $\frac{\Delta x(T)}{\Delta b}$ both very large, a case of the exploding gradient. If $(x(0), b)$ puts the system far from an unstable point, then a small variation in $x(0)$ would have no effect on $x(T)$, making $\frac{\Delta x(T)}{\Delta x(0)} = 0$, a case of the vanishing gradient. Note that in this case, $\frac{\Delta x(T)}{\Delta b} \approx \frac{\partial x(T)}{\partial b} = \frac{1}{x(T)(1-x(T)) - 5} - 1$ neither decays to zero nor blows up to infinity. Indeed, it's the only well-behaved gradient, which explains why early researches focused on learning or designing recurrent networks systems that could perform long-ranged computations (such as outputting the first input it sees at the very end of an episode) by shaping its stable attractors. For the general case, the intuition still holds (Figures 3, 4, and 5).

4.

$$\nabla_x F(x_{t-1}, u_t, \theta) \nabla_x F(x_{t-2}, u_{t-1}, \theta) \cdots \nabla_x F(x_{t-k}, u_{t-k+1}, \theta) = W_{\text{rec}} \text{diag}(\sigma'(x_{t-1})) W_{\text{rec}} \text{diag}(\sigma'(x_{t-2})) \cdots W_{\text{rec}} \text{diag}(\sigma'(x_{t-k}))$$